Lecture 7
Eigenvalues and Eigenvectors

7.1 Eigenvalues and Eigenvectors

7.2 Diagonalization

7.3 Symmetric Matrices and Orthogonal Diagonalization

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7.1 Eigenvalues and Eigenvectors

- Eigenvalue problem:

  If A is an $n \times n$ matrix, do there exist nonzero vectors $\boldsymbol{x}$ in R^n such that $A\boldsymbol{x}$ is a scalar multiple of $\boldsymbol{x}$?

- Eigenvalue and eigenvector:

  A : an $n \times n$ matrix

  λ : a scalar

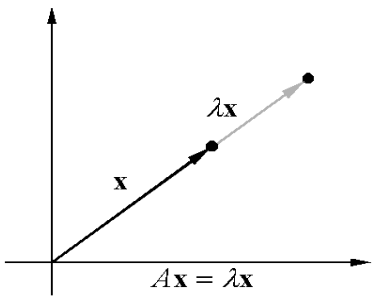
  $\boldsymbol{x}$: a nonzero vector in R^n

  Eigenvalue

  $$A\boldsymbol{x} = \lambda \boldsymbol{x}$$

  Eigenvector

- Geometrical Interpretation



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- Ex 1: (Verifying eigenvalues and eigenvectors)

$$A=\begin{bmatrix}2 & 0\\ 0 & -1\end{bmatrix} \quad x_1=\begin{bmatrix}1\\ 0\end{bmatrix} \quad x_2=\begin{bmatrix}0\\ 1\end{bmatrix}$$

Eigenvalue

$$Ax_1=\begin{bmatrix}2 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix}=\begin{bmatrix}2\\ 0\end{bmatrix}=2\begin{bmatrix}1\\ 0\end{bmatrix}=2x_1$$

Eigenvector

Eigenvalue

$$Ax_2=\begin{bmatrix}2 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix}=\begin{bmatrix}0\\ -1\end{bmatrix}=-1\begin{bmatrix}0\\ 1\end{bmatrix}=(-1)x_2$$

Eigenvector

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- Thm 7.1: (The eigenspace of A corresponding to λ)

If A is an $n \times n$ matrix with an eigenvalue λ, then the set of all eigenvectors of λ together with the zero vector is a subspace of R^n. This subspace is called **the eigenspace of λ** .

Pf:

x_1 and x_2 are eigenvectors corresponding to λ

$(i.e.\ Ax_1=\lambda x_1,\ Ax_2=\lambda x_2)$

(1) $A(x_1+x_2)=Ax_1+Ax_2=\lambda x_1+\lambda x_2=\lambda(x_1+x_2)$

$(i.e.\ x_1+x_2$ is an eigenvector corresponding to $\lambda)$

(2) $A(cx_1)=c(Ax_1)=c(\lambda x_1)=\lambda(cx_1)$

$(i.e.\ cx_1$ is an eigenvector corresponding to $\lambda)$

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- Ex 3: (An example of eigenspaces in the plane)

Find the eigenvalues and corresponding eigenspaces of

$$A=\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}$$

Sol:

If $\mathrm{v}=(x,y)$

$$A\mathrm{v}=\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}-x\\ y\end{bmatrix}$$

For a vector on the x-axis Eigenvalue $\lambda_1=-1$

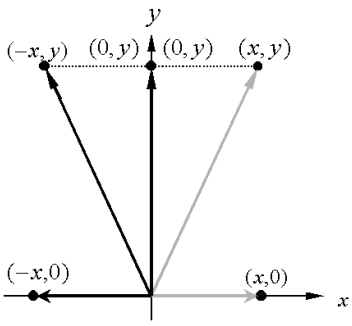
$$\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}x\\ 0\end{bmatrix}=\begin{bmatrix}-x\\ 0\end{bmatrix}=-1\begin{bmatrix}x\\ 0\end{bmatrix}$$

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For a vector on the y-axis Eigenvalue $\lambda_2=1$

$$\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}0\\ y\end{bmatrix}=\begin{bmatrix}0\\ y\end{bmatrix}=1\begin{bmatrix}0\\ y\end{bmatrix}$$

Geometrically, multiplying a vector (x, y) in R^2 by the matrix A corresponds to a reflection in the y-axis.



The eigenspace corresponding to $\lambda_1=-1$ is the x-axis.

The eigenspace corresponding to $\lambda_2=1$ is the y-axis.

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- Thm 7.2: (Finding eigenvalues and eigenvectors of an $n \times n$ matrix A) Let A is an $n \times n$ matrix.

  (1) An eigenvalue of A is a scalar λ such that $\det(\lambda \mathrm{I} - A) = 0$.

  (2) The eigenvectors of A corresponding to λ are the nonzero solutions of $(\lambda \mathrm{I} - A)x = 0$.

- Note:

  $Ax = \lambda x \Rightarrow (\lambda \mathrm{I} - A)x = 0$ (homogeneous system)

  If $(\lambda \mathrm{I} - A)x = 0$ has nonzero solutions iff $\det(\lambda \mathrm{I} - A) = 0$.

- Characteristic polynomial of A:

$$\det(\lambda \mathrm{I} - A) = \left|(\lambda \mathrm{I} - A)\right| = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$$

- Characteristic equation of A:

$$\det(\lambda \mathrm{I} - A) = 0$$

- Ex 4: (Finding eigenvalues and eigenvectors)

$$A = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix}$$

Sol: Characteristic equation:

$$\det(\lambda \mathrm{I} - A) = \begin{vmatrix} \lambda - 2 & 12 \\ -1 & \lambda + 5 \end{vmatrix}$$

$$= \lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2) = 0$$

$$\Rightarrow \lambda = -1, -2$$

Eigenvalues: $\lambda_1 = -1, \lambda_2 = -2$

$$(1)\lambda_1 = -1 \Rightarrow (\lambda_1 \mathrm{I} - A)x = \begin{bmatrix} -3 & 12 \\ -1 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\because \begin{bmatrix} -3 & 12 \\ -1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4t \\ t \end{bmatrix} = t\begin{bmatrix} 4 \\ 1 \end{bmatrix},\ t \neq 0$$

$$(2)\lambda_2 = -2 \Rightarrow (\lambda_2 \mathrm{I} - A)x = \begin{bmatrix} -4 & 12 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\because \begin{bmatrix} -4 & 12 \\ -1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3t \\ t \end{bmatrix} = t\begin{bmatrix} 3 \\ 1 \end{bmatrix},\ t \neq 0$$

Check : $Ax = \lambda_i x$

- Ex 5: (Finding eigenvalues and eigenvectors)

  Find the eigenvalues and corresponding eigenvectors for the matrix A. What is the dimension of the eigenspace of each eigenvalue?

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Sol: Characteristic equation:

$$|\lambda \mathrm{I} - A| = \begin{vmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 2 \end{vmatrix} = (\lambda - 2)^3 = 0$$

Eigenvalue: $\lambda = 2$

The eigenspace of A corresponding to $\lambda = 2$:

$$(\lambda \mathrm{I} - A)x = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\because \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} s \\ 0 \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \ s,t \neq 0$$

$$\left\{ s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \middle| s,t \in R \right\} : \text{the eigenspace of A corresponding to } \lambda = 2$$

Thus, the dimension of its eigenspace is 2.

- Notes:

(1) If an eigenvalue λ_1 occurs as a multiple root (*k times*) for the characteristic polynominal, then we say λ_1 has **multiplicity** *k*.

(2) The multiplicity of an eigenvalue is greater than or equal to the dimension of its eigenspace.

▪ Ex 6：Find the eigenvalues of the matrix A and find a basis for each of the corresponding eigenspaces.

$$A=\begin{bmatrix}1&0&0&0\\0&1&5&-10\\1&0&2&0\\1&0&0&3\end{bmatrix}$$

Sol: Characteristic equation:

$$|\lambda \mathrm{I}-A|=\begin{vmatrix}\lambda-1&0&0&0\\0&\lambda-1&-5&10\\-1&0&\lambda-2&0\\-1&0&0&\lambda-3\end{vmatrix}$$

$$=(\lambda-1)^2(\lambda-2)(\lambda-3)=0$$

Eigenvalues : $\lambda_1=1,\lambda_2=2,\lambda_3=3$

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(1) $\lambda_1=1$

$$\Rightarrow(\lambda_1\mathrm{I}-A)x=\begin{bmatrix}0&0&0&0\\0&0&-5&10\\-1&0&-1&0\\-1&0&0&-2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}0&0&0&0\\0&0&-5&10\\-1&0&-1&0\\-1&0&0&-2\end{bmatrix}\sim\begin{bmatrix}1&0&0&2\\0&0&1&-2\\0&0&0&0\\0&0&0&0\end{bmatrix}\Rightarrow\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}=\begin{bmatrix}-2t\\s\\2t\\t\end{bmatrix}=s\begin{bmatrix}0\\1\\0\\0\end{bmatrix}+t\begin{bmatrix}-2\\0\\2\\1\end{bmatrix},\ s,t\neq 0$$

$$\Rightarrow\left\{\begin{bmatrix}0\\1\\0\\0\end{bmatrix},\begin{bmatrix}-2\\0\\2\\1\end{bmatrix}\right\}$$ is a basis for the eigenspace of A corresponding to $\lambda=1$

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$$(2)\lambda_2 = 2$$

$$\Rightarrow (\lambda_2 \mathrm{I} - A)x = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5 & 10 \\ -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5 & 10 \\ -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 5t \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 0 \\ 5 \\ 1 \\ 0 \end{bmatrix}, \quad t \neq 0$$

$$\Rightarrow \left\{ \begin{bmatrix} 0 \\ 5 \\ 1 \\ 0 \end{bmatrix} \right\}$$ is a basis for the eigenspace of A corresponding to $\lambda = 2$

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$$(3)\lambda_3 = 3$$

$$\Rightarrow (\lambda_3 \mathrm{I} - A)x = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & -5 & 10 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & -5 & 10 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -5t \\ 0 \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \quad t \neq 0$$

$$\Rightarrow \left\{ \begin{bmatrix} 0 \\ -5 \\ 0 \\ 1 \end{bmatrix} \right\}$$ is a basis for the eigenspace of A corresponding to $\lambda = 3$

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- Thm 7.3: (Eigenvalues of triangular matrices)

  If A is an $n \times n$ triangular matrix, then its eigenvalues are the entries on its main diagonal.

- Ex 7: (Finding eigenvalues for diagonal and triangular matrices)

$$(a)A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 5 & 3 & -3 \end{bmatrix} \quad (b)A = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$$

Sol:

$$(a)\ |\lambda \mathrm{I} - A| = \begin{vmatrix} \lambda - 2 & 0 & 0 \\ 1 & \lambda - 1 & 0 \\ -5 & -3 & \lambda + 3 \end{vmatrix} = (\lambda - 2)(\lambda - 1)(\lambda + 3)$$

$$\lambda_1 = 2, \lambda_2 = 1, \lambda_3 = -3$$

$$(b)\ \lambda_1 = -1, \lambda_2 = 2, \lambda_3 = 0, \lambda_4 = -4, \lambda_5 = 3$$

- Eigenvalues and eigenvectors of linear transformations:

A linear transformation $T : V \to V$ can always be represented by a matrix A. So, we can imitate the notion of eigenvectors λ and eigenvector $\mathbf{x}$ of a matrix A for the linear transformation T.

A number λ is called an eigenvalue of a linear transformation $T : V \to V$ if there is a nonzero vector $\mathbf{x}$ such that $T(\mathbf{x}) = \lambda \mathbf{x}$. The vector $\mathbf{x}$ is called an eigenvector of T corresponding to λ, and the set of all eigenvectors of λ (with the zero vector) is called the eigenspace of λ.

- Ex 8: (Eigenvalues and eigenspaces of a transformation)

Consider the linear transformation T from R^3 into R^3 given by

$$T(x_1,x_2,x_3)=(x_1 + 3x_2,\ 3x_1 + x_2,\ -2x_3).$$

Find its eigenvalues and eigenvectors.

Call A the standard matrix of the transformation T. This is the matrix having columns being coordinate columns of $T(e_1)$, $T(e_2)$ and $T(e_3)$. Here $B=\{e_1 = (1,0,0),\ e_2 = (0,1,0),\ e_3 = (0,0,1)\}$ is the standard basis of R^3. This is computed as follows.

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- Ex 8: (Eigenvalues and eigenspaces of a transformation)

Find the eigenvalues and corresponding eigenspaces

$$A=\begin{bmatrix}1 & 3 & 0\\ 3 & 1 & 0\\ 0 & 0 & -2\end{bmatrix}.$$

Sol:

$$|\lambda I - A| = \begin{vmatrix}\lambda-1 & -3 & 0\\ -3 & \lambda-1 & 0\\ 0 & 0 & \lambda+2\end{vmatrix} = (\lambda+2)^2(\lambda-4)$$

$$\text{eigenvalues}: \lambda_1 = 4, \lambda_2 = -2$$

The eigenspaces for these two eigenvalues are as follows.

$B_1 = \{(1,1,0)\}$ Basis for $\lambda_1 = 4$

$B_2 = \{(1,-1,0),(0,0,1)\}$ Basis for $\lambda_2 = -2$

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- Notes:

(1) Let $T:R^3 \to R^3$ be the linear transformation whose standard matrix is A in Ex. 8, and let B' be the basis of R^3 made up of three linear independent eigenvectors found in Ex. 8. Then A', the matrix of T relative to the basis B', is diagonal.

$$B' = \{(1,1,0),(1,-1,0),(0,0,1)\}$$

Eigenvectors of A

$$A' = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Eigenvalues of A

(2) The main diagonal entries of the matrix A' are the eigenvalues of A.

7.2 Diagonalization

- Diagonalization problem:

For a square matrix A, does there exist an invertible matrix P such that $P^{-1}AP$ is diagonal?

- Diagonalizable matrix:

A square matrix A is called **diagonalizable** if there exists an invertible matrix P such that $P^{-1}AP$ is **a diagonal matrix**.

(P diagonalizes A)

- Notes:

(1) If there exists an invertible matrix P such that $B = P^{-1}AP$, then two square matrices A and B are called **similar**.

(2) The eigenvalue problem is related closely to the diagonalization problem.

- Thm 7.4: (Similar matrices have the same eigenvalues)

  If A and B are similar $n \times n$ matrices, then they have the same eigenvalues.

Pf:

A and B are similar $\Rightarrow B = P^{-1}AP$

$$\begin{aligned}|\lambda \mathrm{I} - B| &= |\lambda \mathrm{I} - P^{-1}AP| = |P^{-1}\lambda \mathrm{I}P - P^{-1}AP| = |P^{-1}(\lambda \mathrm{I} - A)P| \\ &= |P^{-1}||\lambda \mathrm{I} - A||P| = |P^{-1}||P||\lambda \mathrm{I} - A| = |P^{-1}P||\lambda \mathrm{I} - A| \\ &= |\lambda \mathrm{I} - A|\end{aligned}$$

A and B have the same characteristic polynomial.
Thus A and B have the same eigenvalues.

- Ex 1: (A diagonalizable matrix)

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Sol: Characteristic equation:

$$|\lambda \mathrm{I} - A| = \begin{vmatrix} \lambda - 1 & -3 & 0 \\ -3 & \lambda - 1 & 0 \\ 0 & 0 & \lambda + 2 \end{vmatrix} = (\lambda - 4)(\lambda + 2)^2 = 0$$

Eigenvalues : $\lambda_1 = 4, \lambda_2 = -2, \lambda_3 = -2$

$$(1)\lambda = 4 \Rightarrow \text{ Eigenvector:} p_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$(2)\lambda=-2 \Rightarrow \text{Eigenvector}: p_2=\begin{bmatrix}1\\-1\\0\end{bmatrix},\ p_3=\begin{bmatrix}0\\0\\1\end{bmatrix}\ \text{(See Ex.5)}$$

$$P=[p_1\quad p_2\quad p_3]=\begin{bmatrix}1&1&0\\1&-1&0\\0&0&1\end{bmatrix}\Rightarrow P^{-1}AP=\begin{bmatrix}4&0&0\\0&-2&0\\0&0&-2\end{bmatrix}$$

- Notes:

$$(1)\ P=[p_2\quad p_1\quad p_3]=\begin{bmatrix}1&1&0\\-1&1&0\\0&0&1\end{bmatrix}\Rightarrow P^{-1}AP=\begin{bmatrix}-2&0&0\\0&4&0\\0&0&-2\end{bmatrix}$$

$$(2)\ P=[p_2\quad p_3\quad p_1]=\begin{bmatrix}1&0&1\\-1&0&1\\0&1&0\end{bmatrix}\Rightarrow P^{-1}AP=\begin{bmatrix}-2&0&0\\0&-2&0\\0&0&4\end{bmatrix}$$

- Thm 7.5: (Condition for diagonalization)

An $n\times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors.

Pf:

$(\Rightarrow)$ A is diagonalizable

there exists an invertible P s.t. $D=P^{-1}AP$ is diagonal

Let $P=[p_1 \mid p_2 \mid \cdots \mid p_n]$ and $D=diag(\lambda_1,\lambda_2,\cdots,\lambda_n)$

$$PD=[p_1 \mid p_2 \mid \cdots \mid p_n]\begin{bmatrix}\lambda_1&0&\cdots&0\\0&\lambda_2&\cdots&0\\\vdots&\vdots&\ddots&\vdots\\0&0&\cdots&\lambda_n\end{bmatrix}$$

$$=[\lambda_1 p_1 \mid \lambda_2 p_2 \mid \cdots \mid \lambda_n p_n]$$

$$AP = A[p_1 \mid p_2 \mid \cdots \mid p_n] = [Ap_1 \mid Ap_2 \mid \cdots \mid Ap_n]$$

$\because AP = PD$

$\therefore Ap_i = \lambda_i p_i,\ i = 1, 2, \ldots, n$

($i.e.$ the column vector p_i of P are eigenvectors of A)

$\because P$ is invertible $\Rightarrow p_1, p_2, \cdots, p_n$ are linearly independent.

$\therefore$ A has n linearly independent eigenvectors.

($\Leftarrow$) A has n linearly independent eigenvectors $p_1, p_2, \cdots p_n$ with corresponding eigenvalues $\lambda_1, \lambda_2, \cdots \lambda_n$

i.e. $Ap_i = \lambda_i p_i,\ i = 1, 2, \ldots, n$

Let $P = [p_1 \mid p_2 \mid \cdots \mid p_n]$

$$\begin{aligned} AP &= A[p_1 \mid p_2 \mid \cdots \mid p_n] \\ &= [Ap_1 \mid Ap_2 \mid \cdots \mid Ap_n] \\ &= [\lambda_1 p_1 \mid \lambda_2 p_2 \mid \cdots \mid \lambda_n p_n] \\ &= [p_1 \mid p_2 \mid \cdots \mid p_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} = PD \end{aligned}$$

$\because p_1, p_1, \cdots, p_n$ are linearly independent $\Rightarrow P$ is invertible

$\therefore P^{-1}AP = D$

$\Rightarrow A$ is diagonalizable

Note: If n linearly independent eigen vectors do not exist, then the $n \times n$ matrix A is not diagonalizable.

- Ex 4: (A matrix that is not diagonalizable)

Show that the following matrix is not diagonalizable.

$$A=\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$$

Sol: Characteristic equation:

$$|\lambda \mathrm{I}-A|=\begin{vmatrix}\lambda-1 & -2\\ 0 & \lambda-1\end{vmatrix}=(\lambda-1)^2=0$$

$$\text{Eigenvalue}:\ \lambda_1=1$$

$$\lambda \mathrm{I}-A=I-A=\begin{bmatrix}0 & -2\\ 0 & 0\end{bmatrix}\sim\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\Rightarrow \text{Eigenvector}:\ p_1=\begin{bmatrix}1\\ 0\end{bmatrix}$$

A does not have two (n=2) linearly independent eigenvectors, so A is not diagonalizable.

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- Steps for diagonalizing an $n\times n$ square matrix:

Step 1: Find n linearly independent eigenvectors $p_1, p_2, \cdots, p_n$ for A with corresponding eigenvalues $\lambda_1, \lambda_2, \cdots, \lambda_n$

Step 2: Let $P=[p_1 \mid p_2 \mid \cdots \mid p_n]$

Step 3:

$$P^{-1}AP=D=\begin{bmatrix}\lambda_1 & 0 & \cdots & 0\\ 0 & \lambda_2 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & \lambda_n\end{bmatrix}, \text{ where } Ap_i=\lambda_i p_i,\ i=1,2,\ldots,n$$

Note:

The order of the eigenvalues used to form P will determine the order in which the eigenvalues appear on the main diagonal of D.

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- Ex 5: (Diagonalizing a matrix)

$$A=\begin{bmatrix}1 & -1 & -1\\ 1 & 3 & 1\\ -3 & 1 & -1\end{bmatrix}$$

Find a matrix P such that $P^{-1}AP$ is diagonal.

Sol: Characteristic equation:

$$|\lambda \mathrm{I}-A|=\begin{vmatrix}\lambda-1 & 1 & 1\\ -1 & \lambda-3 & -1\\ 3 & -1 & \lambda+1\end{vmatrix}=(\lambda-2)(\lambda+2)(\lambda-3)=0$$

Eigenvalues: $\lambda_1=2, \lambda_2=-2, \lambda_3=3$

$\lambda_1=2$

$$\Rightarrow \lambda_1\mathrm{I}-A=\begin{bmatrix}1 & 1 & 1\\ -1 & -1 & -1\\ 3 & -1 & 3\end{bmatrix}\sim\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$\Rightarrow\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}-t\\ 0\\ t\end{bmatrix}=t\begin{bmatrix}-1\\ 0\\ 1\end{bmatrix}\Rightarrow \text{Eigenvector: } p_1=\begin{bmatrix}-1\\ 0\\ 1\end{bmatrix}$$

$\lambda_2=-2$

$$\Rightarrow \lambda_2\mathrm{I}-A=\begin{bmatrix}-3 & 1 & 1\\ -1 & -5 & -1\\ 3 & -1 & -1\end{bmatrix}\sim\begin{bmatrix}1 & 0 & -\frac{1}{4}\\ 0 & 1 & \frac{1}{4}\\ 0 & 0 & 0\end{bmatrix}$$

$$\Rightarrow\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}\frac{1}{4}t\\ -\frac{1}{4}t\\ t\end{bmatrix}=\frac{1}{4}t\begin{bmatrix}1\\ -1\\ 4\end{bmatrix}\Rightarrow \text{Eigenvector: } p_2=\begin{bmatrix}1\\ -1\\ 4\end{bmatrix}$$

$$\lambda_3 = 3 \quad \Rightarrow \lambda_3 \mathrm{I} - A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 3 & -1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \text{Eigenvector}: \; p_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Let } P = [p_1 \quad p_2 \quad p_3] = \begin{bmatrix} -1 & 1 & -1 \\ 0 & -1 & 1 \\ 1 & 4 & 1 \end{bmatrix}$$

$$\Rightarrow \; P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

- Notes: k is a positive integer

$$(1)\; D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} \Rightarrow \; D^k = \begin{bmatrix} d_1^k & 0 & \cdots & 0 \\ 0 & d_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n^k \end{bmatrix}$$

$$(2)\; D = P^{-1}AP$$

$$\Rightarrow \; D^k = (P^{-1}AP)^k$$

$$= (P^{-1}AP)(P^{-1}AP)\cdots(P^{-1}AP)$$

$$= P^{-1}A(PP^{-1})A(PP^{-1})\cdots(PP^{-1})AP$$

$$= P^{-1}AA\cdots AP$$

$$= P^{-1}A^kP$$

$$\therefore A^k = PD^kP^{-1}$$

- Thm 7.6: (Sufficient conditions for diagonalization)

  If an $n \times n$ matrix A has n distinct eigenvalues, then the corresponding eigenvectors are linearly independent and A is diagonalizable.

- Ex 7: (Determining whether a matrix is diagonalizable)

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -3 \end{bmatrix}$$

Sol: Because A is a triangular matrix,

its eigenvalues are the main diagonal entries.

$$\lambda_1 = 1, \lambda_2 = 0, \lambda_3 = -3$$

These three values are distinct, so A is diagonalizable. (Thm.7.6)

Diagonalize a linear transformation: Suppose that $T: V \to V$ is a LT. Find a basis B of V such that the matrix of T relative to B is a diagonal matrix.

- Ex 8: (Finding a diagonalizing matrix for a linear transformation)

  Let $T: R^3 \to R^3$ be the linear transformation given by

  $$T(x_1, x_2, x_3) = (x_1 - x_2 - x_3,\ x_1 + 3x_2 + x_3,\ -3x_1 + x_2 - x_3)$$

  Find a basis B for R^3 such that the matrix for T relative to B is diagonal.

Sol: We need to find a basis of $V=R^3$, consisting of all eigenvectors

The standard matrix for T is given by

$$A = \begin{bmatrix} T(e_1) & T(e_2) & T(e_3) \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{bmatrix}$$

It is the matrix in the Ex. 5. Recall from Ex. 5, characteristic equation of A is

$$|\lambda \mathrm{I} - A| = \begin{vmatrix} \lambda - 1 & 1 & 1 \\ -1 & \lambda - 3 & -1 \\ 3 & -1 & \lambda + 1 \end{vmatrix} = (\lambda - 2)(\lambda + 2)(\lambda - 3) = 0$$

Eigenvalues : $\lambda_1 = 2, \lambda_2 = -2, \lambda_3 = 3$

$$\lambda_1 = 2 \Rightarrow \lambda_1 \mathrm{I} - A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 3 & -1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t \\ 0 \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \text{Eigenvector}: \; p_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda_2 = -2 \Rightarrow \lambda_2 \mathrm{I} - A = \begin{bmatrix} -3 & 1 & 1 \\ -1 & -5 & -1 \\ 3 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{4}t \\ -\frac{1}{4}t \\ t \end{bmatrix} = \frac{1}{4}t \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} \Rightarrow \text{Eigenvector}: \; p_2 = \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

$$\lambda_3 = 3 \Rightarrow \lambda_3 \mathrm{I} - A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 3 & -1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \text{Eigenvector}: \; p_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Eigenvectors $B = \{p_1, p_2, p_3\}$ are linearly independent and hence is a basis for R^3.

Since

$T(p_1)=2.p_1;$ $T(p_2)=(-2)p_2;$ $T(p_3)=3.p_3$

we have

The matrix of T relative to the basis B is $\begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Moreover, matrix $P = [p_1 \quad p_2 \quad p_3] = \begin{pmatrix} -1 & 1 & -1 \\ 0 & -1 & 1 \\ 1 & 4 & 1 \end{pmatrix}$ is the transition matrix from the basis B to the standard basis. So, we get

$$P^{-1}AP = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

7.3 Symmetric Matrices and Orthogonal Diagonalization

- Symmetric matrix:

  A square matrix A is **symmetric** if it is equal to its transpose:

  $$A = A^T$$

- Ex 1: (Symmetric matrices and nonsymmetric matrices)

$$A = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 3 & 0 \\ -2 & 0 & 5 \end{bmatrix}$$ (symmetric)

$$B = \begin{bmatrix} 4 & 3 \\ 3 & 1 \end{bmatrix}$$ (symmetric)

$$C = \begin{bmatrix} 3 & 2 & 1 \\ 1 & -4 & 0 \\ 1 & 0 & 5 \end{bmatrix}$$ (nonsymmetric)

- Thm 7.7: (Eigenvalues of symmetric matrices)

If A is an $n \times n$ symmetric matrix, then the following properties are true.

(1) A is diagonalizable.

(2) All eigenvalues of A are real.

(3) If λ is an eigenvalue of A with multiplicity k, then λ has k linearly independent eigenvectors. That is, the eigenspace of λ has dimension k.

- Ex 2:

Prove that a symmetric matrix is diagonalizable.

$$A = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$$

Pf: Characteristic equation:

$$|\lambda I - A| = \begin{vmatrix} \lambda - a & -c \\ -c & \lambda - b \end{vmatrix} = \lambda^2 - (a+b)\lambda + ab - c^2 = 0$$

As a quadratic in λ, this polynomial has a discriminant of

$$\begin{aligned} (a+b)^2 - 4(ab - c^2) &= a^2 + 2ab + b^2 - 4ab + 4c^2 \\ &= a^2 - 2ab + b^2 + 4c^2 \\ &= (a-b)^2 + 4c^2 \geq 0 \end{aligned}$$

(1) $(a-b)^2+4c^2=0$

$\Rightarrow a=b,\ c=0$

$A=\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ is a matrix of diagonal.

(2) $(a-b)^2+4c^2>0$

The characteristic polynomial of A has two distinct real roots, which implies that A has two distinct real eigenvalues. Thus, A is diagonalizable.

- Orthogonal matrix:

A square matrix P is called **orthogonal** if it is invertible and

$$P^{-1}=P^T$$

- Ex 4: (Orthogonal matrices)

(a) $P=\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ is orthogonal because $P^{-1}=P^T=\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

(b) $P=\begin{bmatrix} \frac{3}{5} & 0 & \frac{-4}{5} \\ 0 & 1 & 0 \\ \frac{4}{5} & 0 & \frac{3}{5} \end{bmatrix}$ is orthogonal because $P^{-1}=P^T=\begin{bmatrix} \frac{3}{5} & 0 & \frac{4}{5} \\ 0 & 1 & 0 \\ \frac{-4}{5} & 0 & \frac{3}{5} \end{bmatrix}$.

- Thm 7.8: (Properties of orthogonal matrices)

  An $n \times n$ matrix P is orthogonal if and only if its column vectors form an orthonormal set.

- Ex 5: (An orthogonal matrix)

$$P = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \\ \frac{-2}{3\sqrt{5}} & \frac{-4}{3\sqrt{5}} & \frac{5}{3\sqrt{5}} \end{bmatrix}$$

Sol: If P is a orthogonal matrix, then $P^{-1} = P^T \Rightarrow PP^T = \mathrm{I}$

$$PP^T = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \\ \frac{-2}{3\sqrt{5}} & \frac{-4}{3\sqrt{5}} & \frac{5}{3\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{-2}{\sqrt{5}} & \frac{-2}{3\sqrt{5}} \\ \frac{2}{3} & \frac{1}{\sqrt{5}} & \frac{-4}{3\sqrt{5}} \\ \frac{2}{3} & 0 & \frac{5}{3\sqrt{5}} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathrm{I}$$

$$\text{Let } p_1 = \begin{bmatrix} \frac{1}{3} \\ \frac{-2}{\sqrt{5}} \\ \frac{-2}{3\sqrt{5}} \end{bmatrix}, p_2 = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{\sqrt{5}} \\ \frac{-4}{3\sqrt{5}} \end{bmatrix}, p_3 = \begin{bmatrix} \frac{2}{3} \\ 0 \\ \frac{5}{3\sqrt{5}} \end{bmatrix}$$

produces

$$p_1 \cdot p_2 = p_1 \cdot p_3 = p_2 \cdot p_3 = 0$$

$$\|p_1\| = \|p_2\| = \|p_3\| = 1$$

$\{p_1, p_2, p_3\}$ is an orthonormal set.

- Thm 7.9: (Properties of symmetric matrices)

  Let A be an $n \times n$ symmetric matrix. If λ_1 and λ_2 are distinct eigenvalues of A, then their corresponding eigenvectors x_1 and x_2 are orthogonal.

- Ex 6: (Eigenvectors of a symmetric matrix)

  Show that any two eigenvectors of $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

  corresponding to distinct eigenvalues are orthogonal

- Sol: Characteristic function

$$|\lambda \mathrm{I} - A| = \begin{vmatrix} \lambda - 3 & -1 \\ -1 & \lambda - 3 \end{vmatrix} = \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4) = 0$$

$$\Rightarrow \text{Eigenvalues}: \ \lambda_1 = 2, \lambda_2 = 4$$

$$(1)\ \lambda_1 = 2 \Rightarrow \lambda_1 \mathrm{I} - A = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow \mathrm{x}_1 = s \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \ s \neq 0$$

$$(2)\ \lambda_2 = 4 \Rightarrow \lambda_2 \mathrm{I} - A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \Rightarrow \mathrm{x}_2 = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \ t \neq 0$$

$$\mathrm{x}_1 \cdot \mathrm{x}_2 = \begin{bmatrix} -s \\ s \end{bmatrix} \cdot \begin{bmatrix} t \\ t \end{bmatrix} = st - st = 0 \Rightarrow \mathrm{x}_1 \text{ and } \mathrm{x}_2 \text{ are orthogonal.}$$

USTH

- Orthogonal Diagonalization:

  A matrix A is **orthogonally diagonalizable** if there exists an orthogonal matrix P such that

$$P^{-1}.A.P = D$$

  is a diagonal matrix.

- Thm 7.10: (Fundamental theorem of symmetric matrices)

  Let A be an $n \times n$ matrix. Then A is orthogonally diagonalizable and has real eigenvalue if and only if A is symmetric.

USTH

- Orthogonal diagonalization of a symmetric matrix:

Let A be an $n \times n$ symmetric matrix.

(1) Find all eigenvalues of A and determine the multiplicity of each.

(2) For each eigenvalue of multiplicity 1, choose a unit eigenvector.

(3) For each eigenvalue of multiplicity $k \geq 2$, find a set of k linearly independent eigenvectors. If this set is not orthonormal, apply Gram-Schmidt orthonormalization process.

(4) The composite of steps 2 and 3 produces an orthonormal set of n eigenvectors. Use these eigenvectors to form the columns of P. The matrix $P^{-1}AP = P^{T}AP = D$ will be diagonal.

USTH

- Ex 7: (Determining whether a matrix is orthogonally diagonalizable)

	Symmetric matrix	Orthogonally diagonalizable
$A_1 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	○	○
$A_2 = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 1 & 8 \\ -1 & 8 & 0 \end{bmatrix}$	×	×
$A_3 = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix}$	×	×
$A_4 = \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}$	○	○

USTH

- Ex 9: (Orthogonal diagonalization)

Find an orthogonal matrix P that diagonalizes A.

$$A=\begin{bmatrix} 2 & 2 & -2 \\ 2 & -1 & 4 \\ -2 & 4 & -1 \end{bmatrix}$$

Sol:

(1) $|\lambda I - A| = (\lambda-3)^2(\lambda+6) = 0$

$\lambda_1 = -6, \lambda_2 = 3$ (has a multiplicity of 2)

(2) $\lambda_1 = -6, \quad v_1 = (1,-2,2) \Rightarrow u_1 = \dfrac{v_1}{\|v_1\|} = (\frac{1}{3}, \frac{-2}{3}, \frac{2}{3})$

(3) $\lambda_2 = 3, \quad v_2 = (2,1,0), \; v_3 = (-2,0,1)$

Linear Independent

Gram-Schmidt Process:

$$w_2 = v_2 = (2,1,0), \quad w_3 = v_3 - \frac{v_3 \cdot w_2}{w_2 \cdot w_2} w_2 = (\tfrac{-2}{5}, \tfrac{4}{5}, 1)$$

$$u_2 = \frac{w_2}{\|w_2\|} = (\tfrac{2}{\sqrt{5}}, \tfrac{1}{\sqrt{5}}, 0), \quad u_3 = \frac{w_3}{\|w_3\|} = (\tfrac{-2}{3\sqrt{5}}, \tfrac{4}{3\sqrt{5}}, \tfrac{5}{3\sqrt{5}})$$

(4) $P = [p_1 \; p_2 \; p_3] = \begin{bmatrix} \frac{1}{3} & \frac{2}{\sqrt{5}} & \frac{-2}{3\sqrt{5}} \\ \frac{-2}{3} & \frac{1}{\sqrt{5}} & \frac{4}{3\sqrt{5}} \\ \frac{2}{3} & 0 & \frac{5}{3\sqrt{5}} \end{bmatrix}$

$$\Rightarrow \quad P^{-1}AP = P^{T}AP = \begin{bmatrix} -6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$